UCAR Intelligence Platform

Complete Role-Based Specification

Comprehensive Access, Functionality, Dependencies & Data Relationship Matrix

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1. Role Hierarchy Overview

UCAR Bureau (National/Central)

- Super Admin (UCAR)
- Department Head (UCAR) × 8 domains
- Staff of Departments (UCAR)

Institution Level (×30+ institutions)

- Super Admin (Institution) — i.e., Dean / Director
- Department Head (Institution) × 8 domains
- Staff of Departments (Institution)
- Teachers
- Students

Federated Principle: UCAR defines the blueprint (schema structure, KPI formulas, compliance rules). Institutions operate autonomously within that blueprint. Data flows upward as aggregates, never raw.

2. Role 1: Super Admin (UCAR Bureau)

2.1 Identity & Authentication

- Superadmin role in Keycloak with the highest privilege tier.
- MFA mandatory (TOTP or hardware key).
- JWT token carries role=superadmin, scope=public_schema, institution_id=ALL.
- Session timeout: 30 minutes of inactivity, forced re-auth for destructive operations.
- All actions logged to audit.event_log with operator identity and IP.

2.2 What They CAN Access

- Full read access to the public schema (aggregated materialized views across all 30+ institutions).
- Full read access to every inst_XXX schema — through aggregated views. Can drill into any institution's full detail by clicking on its card, accessing: all stats, domain-filtered KPIs, and the institution-specific RAG chatbot with cross-domain correlation capabilities.
- When accessing an institution card, the UCAR Super Admin has the full institutional intelligence suite:
  - RAG Chatbot (Institution-Scoped): Natural language queries over that institution's data — e.g., "Compare IT budget execution against student satisfaction scores" or "Why is employability rate low in Arts when partnerships are active?" Grounds answers in retrieved records; no hallucination. Retains session context for follow-up questions.
  - Cross-Domain Correlation Discovery: Can ask the chatbot to find hidden correlations across all 8 domains (e.g., "Is there a link between infrastructure maintenance delays and academic performance?").
  - Anomaly Alerts with Root-Cause Analysis: SHAP-explained alerts in plain language (French primary, Arabic toggle). Example: "Le taux d'abandon est anormalement élevé ce mois-ci (+2.3σ). Cause principale: les résultats aux examens ont chuté de 18%." UCAR Super Admin is the only role that can update the status of anomalies (acknowledged / under investigation / false positive).
- Write access to:
  - public.kpi_definitions — defines and updates formulas that compute every KPI.
  - public.blueprint_templates — defines the federated schema structure all institutions must mirror.
  - public.compliance_rules — thresholds that trigger alerts.
  - public.anomaly_thresholds — configures contamination factor and sensitivity per domain for PyOD.
  - public.system_config — platform-wide settings.
  - audit.event_log — read-only; append-only enforced at Postgres level.
- User management across all institutions.

2.3 What They CANNOT Access

- Cannot modify raw data inside any inst_XXX schema directly.
- Cannot see individual student PII — only aggregated metrics.
- Cannot bypass the bitemporal audit trail.
- Cannot delete audit log entries under any circumstance.
- Cannot impersonate institutional users without an auditable override.

2.4 Functionalities & Operations

- KPI Formula Management: Define, version, and push KPI calculation formulas via dbt models. Formula changes trigger auto-recalculation across all 30+ institutions.
- Blueprint Versioning & Push: Update the federated schema blueprint. A "Pending Sync" banner notifies all institutional admins.
- Traffic-Light Grid: Primary operational view — matrix of 30+ institutions × 8 domains. Each cell is green/yellow/red. Clicking any institution card opens full institutional intelligence (stats, domain KPIs, chatbot, anomaly details).
- Cross-Institution Benchmarking: Compare any KPI across any subset of institutions. Automatically surfaces best-practice institutions.
- Anomaly Status Management: Receive overnight PyOD anomaly alerts. Only this role can mark anomaly status: acknowledged, under investigation, or false positive.
- Automated National Reports: Configure and trigger weekly/monthly/annual national reports (PDF/Excel). Pre-generated on schedule — instant download. Include forecasts with confidence intervals.
- Predictive Dashboard: AutoARIMA forecasts for enrollment, finance, research output, and dropout rates. Forecast horizon: 6–12 months. Includes uncertainty bands and plain-language summaries.
- Chatbot (RAG — National): Query national aggregate data in natural language. Example: "Which 5 institutions have the worst budget execution rate this semester?"
- Compliance Monitoring: View the bitemporal audit log for any institution. Detect data anomalies — see original value, new value, who changed it, and when.
- System Health Dashboard: Monitor platform performance — API latency, ingestion queue depth, database pool utilization, NATS message throughput.

2.5 Outputs They Generate

- Updated KPI formulas → consumed by all inst_XXX schemas via dbt refresh.
- Blueprint updates → trigger schema sync at institutional level.
- Anomaly status updates → visible to UCAR Department Heads and Institution Deans.
- National benchmark reports → available to institutional admins for self-assessment.
- Compliance audit findings → trigger institutional corrective action workflows.

3. Role 2: Department Head (UCAR Bureau)

There are 8 Department Heads at the UCAR level, one per federated domain: Academic, Finance, HR, Research, Infrastructure, Employment, Partnerships, ESG/CSR.

3.1 Identity & Authentication

- Keycloak role: ucar_dept_head with domain-specific scope (e.g., domain=academic).
- MFA mandatory.
- JWT carries role=ucar_dept_head, domain=<specific_domain>, institution_id=ALL.
- Can only see cross-institutional data for their specific domain.

3.2 What They CAN Access

- Read access to public.kpis_aggregate filtered to their domain only.
- Read access to public.anomaly_alerts filtered to their domain (view only — cannot update status).
- Read access to public.blueprint_templates for their domain.
- Propose changes to KPI formulas for their domain (submitted to Super Admin for approval).
- Read access to anonymized institution-level drill-downs within their domain (no student PII).
- Read access to national domain-specific reports.

3.3 What They CANNOT Access

- Cannot see data from other domains.
- Cannot modify KPI formulas directly — they propose, Super Admin approves.
- Cannot access raw inst_XXX schema data.
- Cannot see individual student or staff records.
- Cannot modify blueprint structures.
- Cannot manage users or system configuration.
- Cannot update anomaly status — read only.

3.4 Functionalities & Operations

- Domain-Specific National Dashboard: Focused view of their domain across all 30+ institutions.
- Domain Benchmarking: Rank institutions by their domain's KPIs. Identify top/underperformers. Generate Best Practice recommendations.
- Anomaly Triage (Domain): View anomaly alerts specific to their domain. Write investigation notes (forwarded to UCAR Super Admin for status change).
- KPI Formula Proposals: Draft updated KPI formulas using a no-code formula builder. Submit for Super Admin review with justification notes.
- Domain Reports: Generate domain-specific national reports including trend analysis, forecasts, and institution-level breakdowns.
- Chatbot (Domain-Scoped): Query national data within their domain.
- Policy Compliance Checking: Verify all institutions meet minimum domain standards.
- Inter-Domain Correlation View (read-only): Can request cross-domain correlation reports — requires Super Admin authorization for sensitive queries.

3.5 Outputs They Generate

- Domain benchmark reports → shared with institutional department heads for self-assessment.
- KPI formula change proposals → submitted to Super Admin for approval.
- Investigation notes → forwarded to institutional department heads.
- Policy compliance flags → trigger notifications to non-compliant institutions.

4. Role 3: Staff of Departments (UCAR Bureau)

These are the operational analysts at the UCAR central office — data clerks, administrative assistants, and junior analysts who support the Department Heads and Super Admin.

4.1 Identity & Authentication

- Keycloak role: ucar_staff with domain-specific scope.
- MFA recommended (configurable by Super Admin).
- JWT carries role=ucar_staff, domain=<assigned_domain>, institution_id=ALL.

4.2 What They CAN Access

- Read access to public.kpis_aggregate filtered to their assigned domain(s).
- Read access to national reports within their domain.
- Read access to anomaly alerts within their domain (view only).
- Read access to document repository (national policy documents, templates).
- Write access to:
  - public.report_queue — schedule and configure report generation jobs.
  - public.meeting_notes — document meeting outcomes and action items.

4.3 What They CANNOT Access

- Cannot access any inst_XXX schema directly.
- Cannot modify KPI formulas, blueprints, or compliance rules.
- Cannot manage users.
- Cannot see individual student or staff data.
- Cannot approve or reject anything — all approvals route to Department Head or Super Admin.
- Cannot access domains outside their assignment.
- Cannot update anomaly status.

4.4 Functionalities & Operations

- Report Preparation: Pull data from dashboards, compile draft reports for Department Head review using pre-built report templates.
- Data Quality Monitoring: Monitor incoming data quality scores from institutions. Flag institutions with consistently low data integrity scores for Department Head attention.
- Meeting Coordination: Schedule inter-institutional meetings, generate agenda documents from dashboard data, record action items.

- Dashboard Monitoring: Watch domain dashboards for anomalies or sudden changes. Escalate to Department Head when patterns emerge.
- Document Management: Upload, tag, and organize national policy documents, circulars, and guidelines for distribution to institutions.
- Chatbot (Limited): Can query domain data for report preparation purposes only.

4.5 Outputs They Generate

- Draft reports → reviewed and approved by UCAR Department Head.
- Data quality flags → reviewed by UCAR Department Head.
- Meeting minutes and action items → distributed to relevant stakeholders.

5. Role 4: Super Admin (Institution-Level)

This is the Dean or Director of a single institution. They are the highest authority within their inst_XXX schema.

5.1 Identity & Authentication

- Keycloak role: inst_admin scoped to their specific institution.
- MFA mandatory.
- JWT carries role=inst_admin, institution_id=<specific_id>.
- Row-Level Security (RLS) in PostgreSQL physically restricts all queries to their inst_XXX schema.

5.2 What They CAN Access

- Full read access to their entire inst_XXX schema — all 8 domain tables, all historical data, all audit logs for their institution.
- Full read access to their institution's KPI scores relative to the national average (anonymized).
- Write access to:
  - inst_XXX.system_config — institution-level settings.
  - inst_XXX.user_accounts — manage all institutional users.
  - inst_XXX.approval_queue — approve/reject data commits from department heads before UCAR sync.
  - inst_XXX.report_config — configure which reports auto-generate and on what schedule.
  - inst_XXX.alert_config — customize local anomaly notification thresholds.
- Read access to public.blueprint_templates and public.kpi_definitions.
- Read access to public.benchmark_reports (anonymized cross-institution comparisons).

5.3 What They CANNOT Access

- Cannot access any other institution's schema — enforced by PostgreSQL RLS at the engine level.

- Cannot modify national KPI formulas or blueprint structures.
- Cannot see which specific institution ranks where in national benchmarks (only their own rank + national average).
- Cannot access UCAR-level admin tools.
- Cannot delete audit log entries.
- Cannot bypass the bitemporal audit trail.

5.4 Functionalities & Operations

- Institutional Traffic-Light Dashboard: Single-institution view — 8 domains, each with green/yellow/red status. Click any cell for SHAP root-cause analysis.
- RAG Chatbot (Institutional Scope): Primary tool for strategic decision-making. Natural language queries over the institution's data. Examples:
  - "Compare our IT budget execution against student satisfaction scores."
  - "Why is the employability rate low in the Arts department when partnerships are active?"
  - "What will our dropout rate look like next semester if current trends continue?"
  - Grounds all answers in retrieved database records — no hallucination. Retains session context for follow-up questions.
- Cross-Domain Correlation Discovery: Can ask the chatbot to find hidden correlations across all 8 domains (e.g., "Is there a link between infrastructure maintenance delays and student academic performance?").
- Anomaly Alerts with Root-Cause Analysis: Receive PyOD-generated alerts with SHAP explanations in plain language (French primary, Arabic toggle). Can view anomaly details and root causes. Anomaly status is managed by UCAR Super Admin.
- Automated Institutional Reports: One-click download of pre-generated reports (weekly/monthly/annual) in PDF and Excel formats.
- Data Commit Approval: Review and approve data batches committed by Department Heads before UCAR sync. Side-by-side diff view of what changed.
- Blueprint Sync: When UCAR pushes a new blueprint or KPI formula, the Dean sees a "National Standard Updated — Click to Sync" banner. Human-in-the-loop acceptance preserved.
- User & Access Management: Create and manage all institutional accounts. Assign department heads to domains.
- Strategic Forecasting: AutoARIMA forecasts for enrollment, budget, research, and dropout — with comparison against national forecast.

5.5 Outputs They Generate

- Approved data commits → trigger NATS event → update public.kpis_aggregate materialized view.
- Institutional reports → archived in document repository.
- User account changes → reflected in Keycloak and access policies.
- Strategic decisions → logged in decision audit trail.

6. Role 5: Department Head (Institution-Level)

There are 8 Department Heads per institution, one per federated domain. They are the AI Stewards for data quality — the critical bridge between raw data entry and trusted institutional intelligence.

6.1 Identity & Authentication

- Keycloak role: inst_dept_head scoped to institution_id + domain.
- MFA recommended.
- JWT carries role=inst_dept_head, institution_id=<specific>, domain=<specific>.
- RLS restricts to their institution + their domain tables only.

6.2 What They CAN Access

- Full read/write access to their domain tables within their inst_XXX schema:
  - Academic Head: academic_records, attendance, exam_results.
  - Finance Head: budget, invoices, grants.
  - HR Head: staff_records, teaching_loads, training.
  - Research Head: publications, patents, projects.
  - Infrastructure Head: assets, rooms, maintenance.
  - Employment Head: alumni, career_fairs, job_placements.
  - Partnerships Head: mous, mobility, erasmus.
  - ESG/CSR Head: energy, accessibility, sustainability.
- Read access to their domain's KPIs and comparison to national average.
- Write access to the AI Ingestion Pipeline — primary uploader and validator of legacy data.
- Write access to inst_XXX.data_commits —
  Read access to anomaly alerts for their domain (view only — cannot update status).

6.3 What They CANNOT Access

- Cannot access other domains within their institution (unless Dean explicitly grants cross-domain read).
- Cannot access other institutions' data.
- Cannot modify KPI formulas or blueprint structures.
- Cannot approve their own data commits — must go through the Dean.
- Cannot manage users outside their domain.
- Cannot see the institution-wide traffic-light grid (that's the Dean's view).
- Cannot access the RAG chatbot for cross-domain queries.
- Cannot update anomaly status.

6.4 Functionalities & Operations

- AI-Powered Document Ingestion (Digitalization of the Past):
  - Upload bulk PDFs and Excel files (legacy records: scanned invoices, old grade sheets, research papers, maintenance logs, staff records).
  - Docling OCR pipeline extracts structured data automatically.
  - Real-time progress feedback via WebSocket: "Page 3/12: table extracted, 47 rows found."

  - No-Code Correction UI: Side-by-side view of original scanned document vs. extracted data. Inline corrections logged in audit trail.
  - NATS JetStream task queue processes uploads sequentially to prevent OOM.
- Data Integrity Dashboard: Gauge showing the department's Data Quality Score — completeness, consistency, and timeliness.
- Domain-Specific Analytics:
  - Academic Head: Class performance trends, failure rate anomaly alerts, attendance heatmaps by day/time.
  - Finance Head: Budget consumption status, audit-ready expenditure logs, grant utilization tracking.
  - HR Head: Staff Stability Index, overtime/workload summaries, training completion rates.
  - Research Head: Innovation Pipeline view, publication impact scores, active project tracking.
  - Infrastructure Head: Equipment Downtime analytics, room availability maps, maintenance schedule optimization.
  - Employment Head: Employability Score, time-to-first-job tracking, career fair ROI.
  - Partnerships Head: Active MoU dashboard, student mobility tracking.
  - ESG/CSR Head: Energy consumption tracking, carbon footprint scoring, accessibility compliance.
- Anomaly Alerts (Domain-Specific): Receive alerts when domain KPIs deviate from historical baseline or national norms. Includes SHAP explanation. View-only — status updates handled by UCAR Super Admin.
- Chatbot (Domain-Scoped): Query their domain data.
- Report Generation: Generate domain-specific reports for the Dean using pre-built templates.

6.5 Outputs They Generate

- Data changed created here automatically change in data base
  Data quality scores → visible to Dean and UCAR Bureau.
- Domain reports → consumed by Dean for institutional oversight.
- Corrected OCR extractions → improve the AI model's accuracy over time (feedback loop).

7. Role 6: Staff of Departments (Institution-Level)

These are the administrative clerks, lab assistants, accountants, and support personnel who handle day-to-day data operations within a specific department.

7.1 Identity & Authentication

- Keycloak role: inst_staff scoped to institution_id + domain + optional row-level restrictions.
- Standard password + optional MFA.
- JWT carries role=inst_staff, institution_id=<specific>, domain=<specific>.

7.2 What They CAN Access

- Read/write access to operational tables within their domain, restricted to their functional scope:
  - Finance staff: can enter invoices, log petty cash, submit supply requests — but cannot see budget totals or grant allocations.
  - HR staff: can log staff attendance, process training certificates — but cannot see salary data or recruitment projections.
  - Academic staff (registrar): can update enrollment records, exam schedules — but cannot modify grades (teacher's responsibility).
  - Infrastructure staff: can log maintenance tickets, update equipment inventory — but cannot approve purchase orders.
- Read access to their domain's operational dashboards (limited to their functional area).

7.3 What They CANNOT Access

- Cannot see data from other domains.
- Cannot see other institutions' data.
- Cannot access KPI dashboards or analytics.
- Cannot use the RAG chatbot.
- Cannot upload PDFs to the ingestion pipeline (Department Head responsibility).
- Cannot approve or commit data batches.
- Cannot see salary data, budget totals, or strategic metrics.
- Cannot manage users.
- Cannot export reports.
- Cannot submit inter-department request tickets.

7.4 Functionalities & Operations

- Fast Data Entry: Streamlined forms for high-frequency, low-latency data entry. Mobile-optimized for field use. Examples:
  - Expense reporting: scan a receipt, auto-extract amount/vendor/date, one-tap submit.
  - Supply requests: select from pre-populated catalog, specify quantity, submit with photo.
  - Maintenance logging: scan equipment QR code, select issue type, attach photo, submit.
  - Event logging: record meeting attendance, equipment checkout/return, visitor registration.
- Form-Based Workflows: No free-text data entry where structured forms can replace it. Dropdown menus, date pickers, barcode/QR scanners, photo attachments.
- Status Tracking: View the status of their submitted entries (approved, pending, rejected). Receive push notifications on mobile when status changes.
- Task Inbox: Receive assigned tasks from their Department Head.
- Budget Consumption View (Finance staff only): See remaining budget for their cost center — but not the total departmental budget or allocation details.
- Incident Reporting: Submit and track incident reports (broken equipment, safety issues). Auto-routed to the appropriate Department Head.

7.5 Outputs They Generate

- Raw data entries → consumed by Department Head for review, validation, and commitment.
- Incident reports → trigger workflows for Department Head action.
- Digitized records → enter the ingestion pipeline for AI extraction.
- Attendance/event logs → feed into KPI calculations.

8. Role 7: Teachers

Teachers are high-frequency data producers — they generate the bulk of academic, research, and infrastructure data through daily operations.

8.1 Identity & Authentication

- Keycloak role: teacher scoped to institution_id.
- Standard password + optional MFA.
- JWT carries role=teacher, institution_id=<specific>, department=<specific>.
- RBAC restricted to specific rows in the Academic domain — a teacher sees only their own classes.

8.2 What They CAN Access

- Read/write access to:
  - inst_XXX.attendance — only for their assigned classes. Teachers are responsible for taking attendance for each session.
  - inst_XXX.grades — only for their assigned classes (continuous assessment and exam marks).
  - inst_XXX.syllabus_status — track and update curriculum delivery progress.
  - inst_XXX.infrastructure_tickets — report broken equipment in their classrooms/labs.
  - inst_XXX.teaching_hours — log actual pedagogical hours worked per week.
  - inst_XXX.publications — submit their own research publications (DOI, journal, co-authors).
  - inst_XXX.grants — submit grant applications and track status.
- Read access to:
  - Auto-generated class performance trends (for their classes only).
  - Personal Research Impact score.
  - Personal Overtime/Load summary.
  - National teaching load standards (to compare their workload against norms).
  - Room availability map for scheduling.
  - Institutional announcements and policy updates.

8.3 What They CANNOT Access

- Cannot see other teachers' grades, classes, or performance data.
- Cannot see departmental KPIs, budget data, or strategic metrics.
- Cannot access the RAG chatbot.
- Cannot upload PDFs to the ingestion pipeline.
- Cannot approve or commit data.

- Cannot see student personal information beyond what is necessary for their classes.
- Cannot modify student records outside their assigned classes.
- Cannot access HR data (salaries, recruitment) — they only see their own workload summary.
- Cannot access other institutions' data.

8.4 Functionalities & Operations

- Fast-Entry Academic Portal:
  - Attendance management (Teacher responsibility): Mark attendance per session using the class roster. Supports QR scan confirmation or manual roster checklist. Takes <30 seconds per class. Data feeds directly into Academic KPIs.
  - Grade entry: Structured form with grade scale validation. Self-Correction prompt appears if grade distribution deviates significantly from national norm.
  - Syllabus progress: Mark topics as "covered," "in progress," or "pending."
- Classroom Incident Reporting (Mobile): Tap, select issue type, attach photo, submit. Creates a ticket in the Infrastructure department's queue. Real-time status updates returned.
- Teaching Hours Logging: Log actual hours worked per week. Auto-calculated overtime alerts.
- Research Output Tracking:
  - Submit publications by pasting DOI — system auto-fills title, journal, citation count.
  - Submit grant applications with deadline tracking.
  - Personal Research Impact dashboard: publication count, h-index approximation, active grants.
- Room/Lab Booking: View real-time room availability map. Book labs and study rooms.
- Student Performance Analytics (Own Classes Only): Auto-generated trends — pass rates, attendance-grade correlation, early warning for at-risk students.
- Notification Feed: Receive institutional announcements, blueprint sync alerts, and deadline reminders.

8.5 Outputs They Generate

- Daily attendance records → feed into Academic KPIs (attendance rate).
- Grade entries → feed into Academic KPIs (success rate, dropout prediction).
- Syllabus progress → feed into pedagogical quality metrics.
- Infrastructure tickets → consumed by Infrastructure Department Head.
- Teaching hours → feed into HR KPIs (workload balance, overtime tracking).
- Research publications/grants → feed into Research KPIs.

9. Role 8: Students

Students are mobile-first consumers and qualitative sensors — they consume their own academic data and provide qualitative feedback that enriches institutional intelligence.

9.1 Identity & Authentication

- Stateless JWT authentication with Redis caching to handle 50,000+ concurrent requests.
- Keycloak role: student scoped to institution_id.
- JWT carries role=student, institution_id=<specific>, student_id=<specific>.
- No MFA required (mobile-first UX priority).

9.2 What They CAN Access

- Read access to:
  - Their own grades, attendance, and academic standing — via the "My UCAR" personal dashboard.
  - Their standing relative to the institutional average (anonymized).
  - Public "Investment per Student" financial transparency view.
  - Room/lab availability map for booking.
  - Alumni network and job boards.
  - Career fair listings and internship postings.
  - Erasmus+ and student mobility application portal.
  - Institutional announcements and academic calendar.
  - Personal "Carbon Footprint" score (ESG department).
- Write access to:
  - Feedback forms: campus accessibility, mobility satisfaction, course evaluations, career aspirations.
  - Incident reports: broken facilities, safety concerns.
  - Study-abroad applications (Erasmus+).
  - Career/professional status update: 6-month post-graduation mobile survey (30 seconds, critical for Employability KPIs).
  - Room/lab booking requests.

9.3 What They CANNOT Access

- Cannot see other students' grades, attendance, or personal data.
- Cannot see any departmental KPIs, budgets, or institutional metrics beyond the "Investment per Student" public view.
- Cannot access teacher performance data.
- Cannot access staff or HR data.
- Cannot access the RAG chatbot.
- Cannot modify any academic records (grades, attendance) — they are consumers, not producers of academic data.
- Cannot see which other students are at risk or flagged.
- Cannot access other institutions' data.

9.4 Functionalities & Operations

- "My UCAR" Personal Dashboard (React Native Mobile App):
  - "My Success" KPI card: grade average, attendance rate, progress toward graduation.
  - Semester-by-semester trend visualization.
  - Standing relative to institutional average (anonymized).
  - Upcoming exams, assignment deadlines, and academic calendar integration.

  - Push notifications for grade postings, schedule changes, and institutional announcements.
- Campus Feedback Forms (One-Touch):
  - Rate classroom experience, campus facilities, food services, transport.
  - Report broken equipment or unsafe conditions with photo attachment.
  - Submit accessibility feedback → feeds ESG/CSR metrics.
  - All forms are mobile-optimized and take <60 seconds.
- Career & Employment Hub:
  - Browse alumni network (anonymized salary benchmarks by degree/year).
  - View career fair schedules and register.
  - Update career aspirations and internship preferences.
  - 6-month post-graduation survey: one screen, 4 questions, 30 seconds. Mission-critical for Employability KPI (UCAR's #1 priority).
- Student Mobility Portal: Apply for Erasmus+ and exchange programs. Track application status. Submit mobility satisfaction feedback.
- Room/Lab Booking: Real-time room availability map. Book study rooms and labs directly from the app.
- Personal Carbon Footprint: View estimated carbon footprint based on commute method, campus energy data, and recycling participation. Gamification elements for sustainability.

9.5 Outputs They Generate

- Course feedback → feed into pedagogical quality assessment.
- Campus incident reports → trigger Infrastructure and ESG workflows.
- Career status surveys → feed Employment KPIs (the single most important data point for UCAR).
- Mobility feedback → feed Partnerships KPIs.
- Accessibility reports → feed ESG/CSR KPIs.
- Room booking requests → feed Infrastructure utilization metrics.

10. Cross-Role Commonalities

10.1 Shared Across ALL Roles

- Single Sign-On (SSO) via Keycloak — one login, one identity, one token.
- Bitemporal Audit Trail: Every action by every role is immutably logged with valid_time and system_time. No role can delete or modify audit entries.
- Multilingual Interface: French (primary) with Arabic toggle.
- Push Notifications: All roles receive role-appropriate notifications.
- No-Code Constraint: No user ever writes SQL or queries directly — all interactions are through forms, dashboards, chatbot, or one-touch mobile interfaces.
- Privacy-by-Design: Every role sees only what they are authorized to see. RLS at the PostgreSQL engine level.
- Responsive Design: Web portal for desktop users (admin, heads, staff). React Native mobile app for teachers and students.
- Saved Preferences: Dashboard filters, language choice, and notification preferences persist across sessions.

10.2 Shared Between Administrative Roles (UCAR + Institutional Admins, Department Heads)

- Dashboard Access (scope varies by role and level).
- Anomaly Alert Reception (view-only for all except UCAR Super Admin who can update status).
- Report Generation (scope and template varies).
- Data Export (PDF/Excel — scope varies).
- Approval Workflows (what they can approve differs by role).

10.3 Shared Between Data-Producing Roles (Staff, Teachers)

- Fast Data Entry with form validation and auto-completion.
- Mobile-Optimized Interfaces for field use.
- Self-Correction Prompts when entered data deviates from norms.
- Status Tracking for submitted entries and tickets.
- Task Inbox for assigned work from Department Heads.

11. Special Specifications Per Role

Role	Special Specification
UCAR Super Admin	Only role that can modify KPI formulas and blueprint structures globally. Only role with system health monitoring access. Only role that can force-sync blueprints. Only role that can update anomaly status (acknowledged / under investigation / false positive). Can access full institutional intelligence for any institution by clicking its card (stats, domain KPIs, institution-specific chatbot, cross-domain correlations).
UCAR Dept Head	Domain-scoped cross-institutional view — sees all 30 institutions but only for their domain. Only role at UCAR level that can propose (not approve) KPI changes. Anomaly alerts are read-only.
UCAR Staff	Purely operational — no strategic access, no approvals. Reduces administrative burden on UCAR Department Heads. No inter-department request handling.
Institution Super Admin (Dean)	Only institutional role with full cross-domain visibility. Only role that can approve data commits before UCAR sync. Full access to RAG chatbot for cross-domain strategic queries and anomaly root-cause analysis.
Institution Dept Head	The AI Steward — only role that operates the Docling ingestion pipeline and the OCR Correction UI. Critical quality gate between raw data and trusted institutional records. Anomaly alerts are read-only.
Institution Staff	Most restricted institutional role — functional access only within their cost center. Cannot see KPIs, budgets, or analytics. Handles volume data entry only. No inter-department request tickets.
Teacher	Row-level RBAC — sees only their own classes. Only role responsible for taking and managing attendance. Only role that produces grades daily. Self-Correction prompts on grade entry are unique to this role.

Student	Stateless JWT with Redis caching (50K concurrent users). Primary interface is React Native mobile app. Only role that produces the post-graduation employment survey — the single most valuable data point for UCAR's #1 KPI.

12. Inter-Role Dependencies & Data Flow

12.1 The Data Pipeline (Who Feeds Whom)

- STUDENTS (Feedback Sensors) → Feedback forms, career surveys
- TEACHERS (High-Frequency Producers) → Daily grades, attendance records (teacher-managed), teaching hours, research outputs
- INSTITUTION STAFF (Volume Entry) → Invoices, maintenance logs, event records
- INSTITUTION DEPT HEAD (AI Steward & Validator) → Ingests legacy PDFs/Excel via Docling, corrects OCR errors, validates data quality, commits validated batch
- INSTITUTION SUPER ADMIN / DEAN (Approver) → Reviews & approves data commits; approval triggers NATS event; uses chatbot for strategic cross-domain queries
- PUBLIC SCHEMA MATERIALIZED VIEW (Auto-Sync) → Background worker updates public.kpis_aggregate
- UCAR DEPT HEAD (Domain Analyst) → Reviews domain-specific national KPIs, identifies best/worst performers, proposes KPI formula updates
- UCAR SUPER ADMIN (Strategic Governor) → Approves KPI changes, pushes blueprint updates, manages anomaly statuses, triggers national reports and compliance audits

12.2 Specific Output → Input Dependencies

Producer Role	Output	Consumer Role	How It's Used
Student	Course feedback	Institution Academic Head	Feeds pedagogical quality assessment
Student	Facility incident report	Institution Infrastructure Head	Creates maintenance ticket and updates asset health
Student	Accessibility feedback	Institution ESG Head	Feeds Green Campus and accessibility compliance metrics
Student	6-month employment survey	Institution Employment Head	Feeds Employability KPI — UCAR's #1 priority
Student	Mobility feedback	Institution Partnerships Head	Feeds Partnerships KPIs
Teacher	Attendance records (teacher-managed)	Institution Academic Head	Feeds attendance rate KPI, triggers absenteeism alerts
Teacher	Daily grades	Institution Academic Head	Feeds success rate, failure rate, dropout prediction KPIs
Teacher	Infrastructure ticket	Institution Infrastructure Head	Creates repair/maintenance work order

Teacher	Teaching hours log	Institution HR Head	Feeds workload balance KPI, overtime alerts
Teacher	Publications/DOIs	Institution Research Head	Feeds publication count, research impact score
Institution Staff	Invoice entries	Institution Finance Head	Feeds budget consumption, expenditure tracking
Institution Staff	Equipment inventory	Institution Infrastructure Head	Feeds asset health, utilization metrics
Institution Staff	Training certificates	Institution HR Head	Feeds training completion rate KPI
Institution Dept Head	Committed data batch	Institution Dean	Appears in approval queue for review
Institution Dept Head	Data quality score	Institution Dean	Appears on institutional health dashboard
Institution Dept Head	Domain report	Institution Dean	Feeds cross-domain strategic analysis
Institution Dean	Approved data commit	UCAR Public Schema	Triggers materialized view refresh via NATS event
UCAR Dept Head	KPI formula proposal	UCAR Super Admin	Submitted for approval and global deployment
UCAR Dept Head	Domain benchmark report	Institution Dept Head	Used for self-assessment against national peers
UCAR Dept Head	Policy compliance flag	Institution Dean	Triggers corrective action workflow
UCAR Super Admin	Updated KPI formula	All Institutions (dbt refresh)	All KPIs recalculated across 30+ institutions
UCAR Super Admin	Blueprint update	All Institution Deans	"Pending Sync" banner, human-in-the-loop acceptance
UCAR Super Admin	Anomaly status update	UCAR Dept Heads + Institution Deans	Anomaly marked as acknowledged, under investigation, or false positive
UCAR Super Admin	National report	All Stakeholders	Published to document repository

12.3 Circular Dependencies (Feedback Loops)

- Teacher → Academic Head → Dean → UCAR → Blueprint → Teacher: If national grade standards change (UCAR pushes new formula), the teacher's Self-Correction prompt thresholds update automatically.
- Student survey → Employment Head → UCAR → Policy → Partnerships Head → Student mobility: Student employability data may trigger UCAR to mandate more partnership MoUs, which opens new Erasmus+ opportunities.

- Dept Head OCR correction → AI model improvement → Better future extractions: Every manual correction to a Docling extraction feeds back into improving the extraction model's accuracy.

13. Database Relationship Map

13.1 Schema Architecture

- AUDIT SCHEMA: audit.event_log (APPEND-ONLY, BITEMPORAL) ← Every role writes here (immutable)
- PUBLIC SCHEMA (UCAR Super Admin & Dept Heads READ):
  - public.kpi_definitions ← UCAR SA WRITE
  - public.blueprint_templates ← UCAR SA WRITE
  - public.compliance_rules ← UCAR SA WRITE
  - public.kpis_aggregate (MATERIALIZED VIEW) ← AUTO
  - public.anomaly_alerts ← PyOD WRITE
  - public.anomaly_thresholds ← UCAR SA WRITE
  - public.benchmark_reports ← SYSTEM WRITE
  - public.forecast_results ← AutoARIMA
- INSTITUTION SCHEMAS (inst_001 … inst_NNN): Each contains 8 Domain Tables (academic, finance, hr, research, infrastructure, employment, partnerships, esg_csr) + user management, approvals, and configuration tables.

13.2 Role-to-Schema Access Matrix

Role	audit.event_log	public schema	inst_XXX schema	Scope
UCAR Super Admin	READ	READ + WRITE (config tables)	READ (full institutional detail on drill-down)	All institutions, all domains
UCAR Dept Head	READ (own domain)	READ (own domain)	—	All institutions, single domain
UCAR Staff	READ (own domain)	READ (own domain, limited)	—	All institutions, single domain, no strategy
Inst. Super Admin	READ (own inst.)	READ (benchmarks)	FULL READ + WRITE (config)	Single institution, all domains
Inst. Dept Head	READ (own domain)	READ (own domain benchmarks)	READ + WRITE (own domain)	Single institution, single domain
Inst. Staff	—	—	WRITE (operational rows only)	Single institution, single domain, functional subset
Teacher	—	—	WRITE (own classes + research)	Single institution, own classes

Student	—	—	READ (own records) + WRITE (feedback)	Single institution, own data only

13.3 Row-Level Security (RLS) Enforcement Points

- Institution isolation: PostgreSQL RLS on institution_id — enforced at engine level. A token with institution_id=7 physically cannot read rows from institution 12.
- Domain isolation: RLS on domain column within institution schemas — Department Head with domain=finance cannot read domain=academic rows.
- Teacher isolation: RLS on teacher_id + class_id — teacher sees only their assigned classes.
- Student isolation: RLS on student_id — student sees only their own records.
- Staff functional scoping: RLS on cost_center or functional_area — staff sees only their operational subset.

This specification ensures that the UCAR Intelligence Platform is not just a database with dashboards — it is a complete operational ecosystem where every role's output is another role's input, every data point flows toward actionable KPIs, and every interaction is designed to eliminate manual work, paper processes, and information silos.

The federated sovereignty model guarantees that institutional autonomy is preserved while national standards are enforced — making the system deployable, scalable, and politically acceptable across all 30+ University of Carthage institutions.